

# The Double-Cycle Geometric Cloud

## An Exact Finite Operator Model for the RH-80 Moving Factor

Liang Wang

July 2026

### Abstract

RH-176 identifies the reduced determinant of a length- $L$  cycle with the geometric section  $\Pi_{L-1}$ . The moving factor required by RH-80 is the square

$$C_N(w) = \Pi_N(w/\lambda)^2, \quad \lambda = 1.678573510428322\dots$$

We give an exact finite normal operator realizing this factor. Let  $L = N + 1$ , restrict the cycle  $C_L$  to the zero-mean subspace, scale by  $\lambda^{-1}$ , and take two copies:

$$K_N = \lambda^{-1}(C_L^\circ \oplus C_L^\circ).$$

Then

$$\det(I - wK_N) = \Pi_N(w/\lambda)^2 = C_N(w).$$

The construction has exactly  $2N$  nonstationary modes, matching the cloud count. Its complete periodic trace ledger is

$$\mathrm{Tr} K_N^m = \frac{2}{\lambda^m} \begin{cases} N, & N+1 \mid m, \\ -1, & N+1 \nmid m. \end{cases}$$

Consequently the RH-80 geometric factor possesses a strict intrinsic finite-orbit representation, not merely a formal polynomial identity.

We also recover the local scattering profile  $(N+1)^{-1}\Pi_N(e^{s/(N+1)}) \rightarrow (e^s - 1)/s$ . A 192-case determinant audit and 104 trace checks have maximum errors  $9.48 \times 10^{-15}$  and below  $10^{-10}$ , respectively. This closes the algebraic model factor only. It does not identify the actual noisy cloud with  $K_N$ , select  $N(\sigma)$ , construct a physical Riesz projection, or prove complement convergence.

## 1 The algebraic gap left by RH-80

RH-80 proves that fixed limiting pole cancellation cannot cross the pole circle. Its correct sufficient replacement is division by the actual finite moving cloud polynomial [4]. In the canonical model that factor is

$$C_N(w) = \Pi_N(w/\lambda)^2, \quad \Pi_N(q) = 1 + q + \dots + q^N. \tag{1}$$

The theorem in RH-80 treats  $C_N$  as a finite determinant once a reducing cloud exists, but it does not specify a canonical finite operator whose determinant is (1).

RH-176 supplies one copy: a reduced cycle of length  $N+1$  has determinant  $\Pi_N$  [3]. Two copies therefore appear inevitable. The point of this paper is to make that observation operator-theoretic and to record the resulting traces, multiplicities, and limitations.

## 2 Definition of the double-cycle cloud

Let  $L = N + 1$ , let  $C_L$  be the forward cycle on  $\mathbb{C}^L$ , and let  $\mathcal{Z}_L = \mathbf{1}_L^\perp$ . Write

$$C_L^\circ = C_L|_{\mathcal{Z}_L}.$$

Its eigenvalues are the nontrivial roots

$$\omega_k = e^{2\pi i k/L}, \quad 1 \leq k \leq N.$$

Define the normal  $2N$ -dimensional operator

$$K_N = \lambda^{-1}(C_L^\circ \oplus C_L^\circ). \quad (2)$$

**Theorem 2.1** (Exact double-cycle realization). *For every  $w \in \mathbb{C}$ ,*

$$\det(I - wK_N) = \prod_{k=1}^N (1 - w\omega_k/\lambda)^2 = \Pi_N(w/\lambda)^2. \quad (3)$$

*Thus  $K_N$  realizes the exact RH-80 geometric moving factor.*

*Proof.* Apply the reduced-cycle determinant identity to each direct-sum copy:

$$\det(I - w\lambda^{-1}C_L^\circ) = \frac{1 - (w/\lambda)^L}{1 - w/\lambda} = \Pi_N(w/\lambda).$$

Determinants multiply over finite invariant direct sums, proving (3).  $\square$

**Corollary 2.2** (Exact dimension dictionary). *The model has cycle length  $L = N + 1$ , reduced rank  $N$  per copy, total cloud rank  $2N$ , and determinant degree  $2N$ .*

This agrees with the RH-15 convention that an effective degree  $N$  selects  $2N$  parity-resolved cloud eigenvalues [2]. Agreement of counts does not identify the observed eigenvalues with the exact roots.

## 3 Complete periodic trace ledger

The cycle powers have

$$\mathrm{Tr} C_L^m = \begin{cases} L, & L \mid m, \\ 0, & L \nmid m. \end{cases}$$

Removing the stationary mode subtracts one from every power trace.

**Theorem 3.1** (Double-cycle trace formula). *For every integer  $m \geq 1$ ,*

$$\mathrm{Tr} K_N^m = \frac{2}{\lambda^m} \begin{cases} N, & N + 1 \mid m, \\ -1, & N + 1 \nmid m. \end{cases} \quad (4)$$

*Equivalently,*

$$\log \det(I - wK_N) = - \sum_{m \geq 1} \frac{w^m}{m} \mathrm{Tr} K_N^m \quad (5)$$

*for  $|w| < \lambda$ .*

*Proof.* On one reduced cycle,

$$\mathrm{Tr}(C_L^o)^m = \mathrm{Tr} C_L^m - 1,$$

which is  $N$  when  $L \mid m$  and  $-1$  otherwise. Scaling contributes  $\lambda^{-m}$  and doubling contributes the factor two. The finite log-determinant expansion is standard inside the reciprocal spectral radius [1].  $\square$

The negative background value  $-2\lambda^{-m}$  is not an error. It is the trace contribution of deleting two stationary modes. At multiples of the cycle length, all phases align and the trace jumps to  $2N\lambda^{-m}$ .

## 4 Coefficient ledger of the squared geometric section

Writing

$$\Pi_N(q)^2 = \sum_{j=0}^{2N} c_{N,j} q^j, \quad (6)$$

the coefficients count pairs  $(a, b)$  with  $0 \leq a, b \leq N$  and  $a + b = j$ .

**Proposition 4.1** (Triangular coefficient law). *The exact coefficients are*

$$c_{N,j} = \begin{cases} j + 1, & 0 \leq j \leq N, \\ 2N - j + 1, & N < j \leq 2N. \end{cases} \quad (7)$$

Therefore

$$\det(I - wK_N) = \sum_{j=0}^{2N} c_{N,j} \lambda^{-j} w^j. \quad (8)$$

*Proof.* For  $j \leq N$ , the admissible pairs are  $(0, j), (1, j - 1), \dots, (j, 0)$ , giving  $j + 1$ . For  $j > N$ , the square  $[0, N]^2$  truncates both ends and leaves  $2N - j + 1$  pairs. Substitute  $q = w/\lambda$ .  $\square$

The coefficient law gives a finite target for the physical Q interface. A future Riesz cloud need not match individual exact roots before one can test its determinant: its normalized coefficients can first be compared with the triangular ledger (8). Newton identities make this equivalent to matching the power traces in Theorem 3.1, but the two diagnostics have different numerical conditioning.

## 5 One-step square roots and the two-step double cycle

RH-15 uses the one-step cloud

$$\mu_{N,k}^{\pm} = \lambda^{-1/2} \exp\left(\pm \frac{ik\pi}{N+1}\right), \quad 1 \leq k \leq N. \quad (9)$$

There are  $2N$  roots and their regularized determinant is  $\Pi_N(z^2/\lambda)$  [2].

**Theorem 5.1** (Squaring produces the double-cycle multiset). *The multiset of squares of the one-step roots (9) is exactly the spectrum of  $K_N$ :*

$$\{(\mu_{N,k}^\pm)^2 : 1 \leq k \leq N\} = \biguplus_{j=1}^N \left\{ \lambda^{-1} e^{2\pi i j / (N+1)}, \lambda^{-1} e^{2\pi i j / (N+1)} \right\} \quad (10)$$

*with every nontrivial root appearing twice.*

*Proof.* Squaring gives  $\lambda^{-1} e^{\pm 2\pi i k / (N+1)}$ . As  $k$  runs from one to  $N$ , the plus sign lists all nontrivial roots once, and the minus sign lists the same set in reverse order. Hence each root has multiplicity two.  $\square$

This theorem supplies an exact algebraic adapter between the RH-15 one-step cloud and the RH-80 two-step squared factor. It is consistent with the general identity relating a one-step regularized determinant to its symmetric two-step Fredholm completion. It still does not prove that the physical one-step resonances square without leakage into the selected physical two-step Riesz cloud; that is an operator-selection problem, not a polynomial identity.

## 6 Normality, uniqueness, and orientation blindness

The operator  $K_N$  is normal. Among finite normal operators, its determinant fixes the multiset of eigenvalues and therefore fixes it up to unitary equivalence. This is a useful limited canonicity: once  $N$ ,  $\lambda$ , and the squared geometric factor are specified, no further normal spectral data are missing.

However, orientation is not spectral data. Replacing either cycle by its inverse conjugates and permutes the root set, leaving (3) and all ordinary traces unchanged. The same scalar cloud factor can therefore arise from

$$C_L^\circ \oplus C_L^\circ, \quad C_L^\circ \oplus (C_L^\circ)^{-1}, \quad \text{or} \quad (C_L^\circ)^{-1} \oplus (C_L^\circ)^{-1}.$$

These completions carry different time arrows but the same determinant. RH-178 adds a rank-one directed marker to separate them.

## 7 Scattering coordinate

The finite section has a nontrivial edge scaling. Put  $q = e^{s/(N+1)}$ . Then

$$\frac{1}{N+1} \Pi_N(q) = \frac{e^s - 1}{(N+1)(e^{s/(N+1)} - 1)}. \quad (11)$$

**Proposition 7.1** (Geometric scattering profile). *For every  $s \in \mathbb{C}$ ,*

$$\frac{1}{N+1} \Pi_N(e^{s/(N+1)}) \longrightarrow \frac{e^s - 1}{s}, \quad (12)$$

*with the removable value one at  $s = 0$ , locally uniformly in  $s$ . The normalized double-cycle determinant profile is the square of this limit.*

*Proof.* The denominator in (11) converges locally uniformly to  $s$ . The removable value follows by differentiation or continuity. Squaring preserves locally uniform convergence on compact sets.  $\square$

This is the same canonical profile used in the RH-15 scattering analysis. Here it arises from a literal finite normal operator, not only from a polynomial approximation.

## 8 Audit

The audit uses degrees

$$N = 1, 2, 3, 4, 7, 12, 20, 32$$

and 24 random complex spectral parameters per degree inside  $|w| < 0.8\lambda$ , giving 192 determinant cases. The maximum relative error between the eigenvalue product and  $\Pi_N(w/\lambda)^2$  is  $9.47 \times 10^{-15}$ .

For each degree it also checks powers up to  $\min(2(N+1), 20)$ , for 104 trace cases. The maximum absolute difference from (4) is below  $10^{-10}$ . Finally, the five edge coordinates  $s = -1, -1/2, 0, 1/2, 1$  are evaluated; at  $N = 32$  the finite profile is already close to  $(e^s - 1)/s$ . These are implementation checks of exact finite identities.

## 9 Relation to the physical Gate-A program

The theorem closes an algebraic leaf that may be denoted  $Q_{\text{geom}}$ :

squared geometric cloud factor  $\iff$  double reduced cycle determinant.

It does not close the physical cloud ledger  $Q_{\text{phys}}$ . The latter must prove that the actual noisy Riesz cloud has the same divisor or converges to it in the required coefficient sense. Nor does this paper provide the physical packet-to-Riesz map  $R$ , the complementary Schatten limit  $U$ , the normalization leaf  $Z$ , or the directed limit  $T$ .

The next two concrete tasks are:

1. supplement the scalar determinant with orientation-separating marked traces;
2. compare  $L = N + 1$  with the reset clock rank and observed cloud degree.

No self-adjoint Hilbert–Polya operator, prime-power trace formula, zeta divisor identity, or Riemann Hypothesis is claimed.

## References

- [1] Barry Simon. *Trace Ideals and Their Applications*. American Mathematical Society, 2005.
- [2] Liang Wang. Parity-extracted bulk scattering, 2026. Prime Dynamics Theory Program, RH-15.
- [3] Liang Wang. Finite cyclic closure of normalized history, 2026. Prime Dynamics Theory Program, RH-176.

- [4] Liang Wang. Moving-cloud renormalization and the relative determinant gate, 2026. Prime Dynamics Theory Program, RH-80.